



Data Dictionary

This data dictionary presents the relational database structure shown in the CrayCare Entity-Relationship Diagram (ERD). Each table lists its exact table name and column names, data type, description, and the format or example value used by the current database.

The table and column names are written exactly as defined in the ERD. Primary keys identify each record, while foreign keys connect related records such as users, tanks, batches, sensor data, feeding activities, and notifications.

Table: users

Field Name	Data Type	Description	Format / Example
uid	varchar	Firebase Auth UID. Primary key of the table.	aB3dE6fG9hJ2kL5mN8pQ1rS4tU7v
full_name	varchar	Complete name of the user.	Juan Dela Cruz
email	varchar	Email address used for sign-in and account recovery.	juan@example.com
role	varchar	owner admin.	owner
status	varchar	active disabled.	active
tank_id	varchar	owned tank (null for admin). References tanks.tank_id.	tank_001
photo_url	text	Link to the user's profile image.	https://example.com/profile.jpg
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: notification_settings

Field Name	Data Type	Description	Format / Example
uid	varchar	one row per user. Primary key of the table. References users.uid.	aB3dE6fG9hJ2kL5mN8pQ1rS4tU7v
sound	boolean	Stores the sound.	true
vibration	boolean	Stores the vibration.	true
critical	boolean	Stores the critical.	true
warning	boolean	Stores the warning.	true
feeding	boolean	Stores the feeding.	true
sampling	boolean	Stores the sampling.	true
operational	boolean	Push preference for automatic actuator and sensor-recovery updates.	true
updated_at	timestamp	Date and time when the record was last updated.	2026-08-31 14:30:00



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Table: hardware_assignments

Field Name	Data Type	Description	Format / Example
id	varchar	always "currentOwner" — single Firestore doc. Primary key of the table.	currentOwner
owner_uid	varchar	Unique identifier of the tank owner. References users.uid.	aB3dE6fG9hJ2kL5mN8 pQ1rS4tU7v
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
assigned_by	varchar	admin who assigned. References users.uid.	admin_uid_001
assigned_at	timestamp	Date and time when the hardware was assigned.	2026-08-31 14:30:00

Table: tanks

Field Name	Data Type	Description	Format / Example
tank_id	varchar	== owner uid by default (1 user = 1 tank). Primary key of the table.	tank_001
owner_uid	varchar	Unique identifier of the tank owner. References users.uid.	aB3dE6fG9hJ2kL5mN8 pQ1rS4tU7v
current_batch_id	varchar	nullable. References batches.batch_id.	Sample text
stocking_date	timestamp	Date when the production batch was stocked.	2026-08-31 14:30:00
last_sample_date	timestamp	Stores the last sample date.	2026-08-31 14:30:00
sample_count	int	Number of valid readings included for sample.	20
initial_population	int	Stores the initial population.	20
initial_total_sample_weight	decimal(10,2)	Stores the initial total sample weight.	25.50
initial_total_sample_length	decimal(10,2)	Stores the initial total sample length.	25.50
is_initialized	boolean	Shows whether initialized is true or false.	true
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: sensor_thresholds

Field Name	Data Type	Description	Format / Example
id	varchar	sensor_name (temperature ph_level dissolved_oxygen turbidity water_level feed_level). Primary key of the table.	temperature
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
sensor_name	varchar	temperature ph_level dissolved_oxygen turbidity water_level feed_level.	temperature
min_value	decimal(10,2)	Lowest value of the configured normal range.	24.00
max_value	decimal(10,2)	Highest value of the configured normal range.	30.00
critical_value	decimal(10,2)	Feed level only: critical-below percentage.	10.00



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Field Name	Data Type	Description	Format / Example
hopper_capacity_grams	decimal(10,2)	Feed level only: calibrated full capacity.	1000.00
updated_at	timestamp	Date and time when the record was last updated.	2026-08-31 14:30:00

Table: sensor_readings

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
temperature	decimal(6,2)	Current water temperature reading.	27.20
ph_level	decimal(5,2)	Current pH reading of the water.	7.30
dissolved_oxygen	decimal(5,2)	Current dissolved oxygen reading.	5.60
turbidity	decimal(8,2)	Current water turbidity reading.	31.00
water_level	decimal(6,2)	Current pond water-level reading.	18.00
feed_level	decimal(5,2)	Hopper percentage; operational, not a WQA feature.	68.00
estimated_feed_grams	decimal(10,2)	Estimate calculated from level percentage and configured hopper capacity.	680.00
recorded_at	timestamp	Date and time when the sensor data was recorded.	2026-08-31 14:30:00

Table: sensor_readings_history

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
temp_min	decimal(6,2)	Lowest recorded temp value.	27.20
temp_max	decimal(6,2)	Highest recorded temp value.	27.20
temp_avg	decimal(6,2)	Average recorded temp value.	27.20
pH_min	decimal(5,2)	Lowest recorded pH value.	7.30
pH_max	decimal(5,2)	Highest recorded pH value.	7.30
pH_avg	decimal(5,2)	Average recorded pH value.	7.30
DO_min	decimal(5,2)	Lowest recorded dissolved oxygen value.	5.60
DO_max	decimal(5,2)	Highest recorded dissolved oxygen value.	5.60
DO_avg	decimal(5,2)	Average recorded dissolved oxygen value.	5.60
turbidity_min	decimal(8,2)	Lowest recorded turbidity value.	31.00
turbidity_max	decimal(8,2)	Highest recorded turbidity value.	31.00
turbidity_avg	decimal(8,2)	Average recorded turbidity value.	31.00



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Field Name	Data Type	Description	Format / Example
waterLevel_min	decimal(6,2)	Lowest recorded water level value.	18.00
waterLevel_max	decimal(6,2)	Highest recorded water level value.	18.00
waterLevel_avg	decimal(6,2)	Average recorded water level value.	18.00
feed_level	decimal(5,2)	Operational snapshot; excluded from WQA inputs.	68.00
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	680.00
recorded_at	timestamp	Date and time when the sensor data was recorded.	2026-08-31 14:30:00

Table: actuators

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
device_id	varchar	pump aerator1 aerator2.	pump
control_mode	varchar	on off auto (requested by owner).	auto
current_state	varchar	on off (actual relay state from ESP32).	on
last_changed	timestamp	Stores the last changed.	2026-08-31 14:30:00

Table: actuator_logs

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
actuator_type	varchar	pump aerator1 aerator2.	Sample text
action	varchar	Description of the action that occurred.	Sample text
type	varchar	on off auto.	auto
logged_at	timestamp	Date and time when the event was recorded in the log.	2026-08-31 14:30:00

Table: feeder_status

Field Name	Data Type	Description	Format / Example
tank_id	varchar	Unique identifier of the related tank. Primary key of the table. References tanks.tank_id.	tank_001
command_id	varchar	Optional manual request correlation; empty for scheduled feeds.	command_001
status_reason	varchar	Human-readable block reason.	Sample text



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Field Name	Data Type	Description	Format / Example
status	varchar	idle checking_feed_level dispensing completed skipped_insufficient blocked.	completed
dispense_count	int	completed feeds (Firestore: dispenseCount).	20
last_seen	timestamp	Firestore: lastSeen.	2026-08-31 14:30:00
last_dispensed_at	timestamp	Stores the last dispensed at.	2026-08-31 14:30:00
last_dispensed_grams	decimal(8,2)	Stores the last dispensed grams.	25.50
feed_level	decimal(5,2)	Estimated percentage of feed remaining in the hopper.	68.00
estimated_feed_grams	decimal(10,2)	Estimated amount of feed remaining in the hopper.	680.00

Table: feeder_schedules

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	schedule_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
time	varchar	HH:MM.	06:00
ampm	varchar	AM PM.	AM
time_value	int	minutes since midnight.	360
grams	decimal(8,2)	Optional; default 20 g. Supported: 20–200 g, multiples of 20. Servo-cycle estimate, not measured weight.	20.00
days	varchar	day-of-week mask "1111111" (Sunday first, 1=active).	1111111
enabled	boolean	Stores the enabled.	true
is_done	boolean	Shows whether done is true or false.	false
effective_at_ms	bigint	UTC time stored as milliseconds; creation/edit/re-enable instant.	1788167400000
last_outcome	varchar	completed skipped_insufficient blocked failed; set from device outcome log.	Sample text
last_occurrence_at	bigint	UTC time stored as milliseconds of the scheduled minute; outcomes are date scoped, no midnight reset.	1788167400000
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: feeder_logs

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	feedlog_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001



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Field Name	Data Type	Description	Format / Example
command_id	varchar	Optional originating manual command id; commands are deleted after durable intent.	command_001
action	varchar	Description of the action that occurred.	Sample text
type	varchar	auto manual missed error.	auto
status	varchar	completed skipped_insufficient blocked failed.	completed
schedule_key	varchar	Optional originating schedule document id; not a required FK because schedules may be deleted.	schedule_001
schedule_time	varchar	Optional originating schedule time, e.g. 4:00 PM.	Sample text
occurrence_at	bigint	Original UTC milliseconds since January 1, 1970, retained through offline retries.	1788167400000
requested_grams	decimal(8,2)	Stores the requested grams.	20.00
estimated_dispensed_grams	decimal(8,2)	Completed cycles only; not scale-measured mass.	20.00
amount_basis	varchar	servo_cycle_estimate.	servo_cycle_estimate
estimated_available_grams	decimal(10,2)	Stores the estimated available grams.	680.00
feed_level_before	decimal(5,2)	Stores the feed level before.	68.00
feed_level_after	decimal(5,2)	Stores the feed level after.	66.00
level_change_detected	boolean	Supporting indication only; not proof of exact dispensed mass.	true
verification_note	varchar	Stores the verification note.	Sample text
logged_at	timestamp	Date and time when the event was recorded in the log.	2026-08-31 14:30:00

Table: feeder_commands

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	command_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
command_type	varchar	feed_now (manual feeding command).	feed_now
grams	decimal(8,2)	optional; 20.0 default when absent.	20.00
issued_by	varchar	Stores the issued by. References users.uid.	aB3dE6fG9hJ2kL5mN8pQ1rS4tU7v
issued_at	timestamp	Date and time when the command was created.	2026-08-31 14:30:00
expires_at	timestamp	App request deadline; ESP also enforces at most 60 seconds after issued_at.	2026-08-31 14:30:00



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Table: feeder_notification_receipts

Field Name	Data Type	Description	Format / Example
log_id	varchar	tanks/{tankId}/feeder_notification_receipts/{logId}. Primary key of the table.	feedlog_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
uid	varchar	Unique identifier of the user account. References users.uid.	aB3dE6fG9hJ2kL5mN8 pQ1rS4tU7v
push_attempt_claimed_at	timestamp	Server-only at-most-once claim, not proof of FCM delivery.	2026-08-31 14:30:00

Table: sensor_ingestion

Field Name	Data Type	Description	Format / Example
id	varchar	Logical staging: sensorIngestion/current or current/history/{docId}. Primary key of the table.	current
hardware_id	varchar	Firestore field: hardwareId.	Sample text
source_tank_id	varchar	Stores the source tank id.	Sample text
source_owner_uid	varchar	Stores the source owner uid.	Sample text
source_assignment_at_ms	bigint	Capture-time assignment identity, not upload-time ownership.	1788167400000
captured_at_ms	bigint	Stores the captured at ms.	1788167400000
routing_status	varchar	quarantined when history cannot be safely assigned.	quarantined
routing_reason	varchar	Stores the routing reason.	Sample text

Table: machine_learning_assessments

Field Name	Data Type	Description	Format / Example
id	varchar	sortable UTC timestamp doc id. Primary key of the table.	20260831T143000
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
level	varchar	Good Moderate Poor Critical Insufficient.	Moderate
model_level	varchar	Raw ML class: Good Moderate Poor Critical.	Moderate
rule_level	varchar	Deterministic rolling-risk safety floor.	Sample text
safety_override	boolean	Stores the safety override.	false
source	varchar	Model version Rule-based fallback Insufficient data Stale sensor history.	Sample text
model_algorithm	varchar	RandomForestClassifier Rule-based Not applied.	Sample text
model_version	varchar	Version embedded in the deployed model artifact.	Sample text



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Field Name	Data Type	Description	Format / Example
training_data_origin	varchar	synthetic during prototype; field when retrained from actual pond history.	Sample text
label_origin	varchar	deterministic_threshold_assessment until independent expert labels are available.	Sample text
model_feature_count	int	Number of valid readings included for model feature.	20
analysis_window_minutes	int	Stores the analysis window minutes.	20
data_status	varchar	ready insufficient stale.	ready
source_recorded_at	timestamp	Latest actual source reading; ISO-8601 in Firestore, nullable.	2026-08-31 14:30:00
source_age_seconds	int	Stores the source age seconds.	20
confidence	int	Stores the confidence.	86
driver	varchar	DO pH temp turbidity waterLevel overall.	DO
driver_label	varchar	Stores the driver label.	Sample text
driver_value	decimal(10,2)	Stores the driver value.	25.50
driver_unit	varchar	Stores the driver unit.	Sample text
driver_min	decimal(10,2)	Lowest recorded driver value.	25.50
driver_max	decimal(10,2)	Highest recorded driver value.	25.50
problem	varchar	Stores the problem.	Sample text
insight	text	Stores the insight.	Sample text
action	text	Description of the action that occurred.	Sample text
ts_epoch	bigint	Stores the ts epoch.	1788167400
timestamp	timestamp	Stores the timestamp.	2026-08-31 14:30:00

Table: batches

Field Name	Data Type	Description	Format / Example
batch_id	varchar	Unique identifier of the related production batch. Primary key of the table.	batch_2026_001
tank_id	varchar	Unique identifier of the related tank. References tanks.tank_id.	tank_001
batch_status	varchar	active harvested superseded.	active
stocking_date	timestamp	wall-clock date/time the batch was stocked. Calendar-day based, NOT a 24-hour rolling clock — setup day = Day 0, next calendar day = Day 1 even if stocked late at night.	2026-08-31 14:30:00
harvest_date	timestamp	Date when harvesting was performed.	2026-08-31 14:30:00
initial_count	int	Number of valid readings included for initial.	20
current_count	int	Number of valid readings included for current.	20
harvest_count	int	Number of valid readings included for harvest.	20



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Field Name	Data Type	Description	Format / Example
total_mortality	int	Stores the total mortality.	20
harvest_weight_grams	decimal(10,2)	Stores the harvest weight grams.	5000.00
initial_abw	decimal(8,2)	Stores the initial abw.	25.50
initial_abl	decimal(8,2)	Stores the initial abl.	25.50
final_abw	decimal(8,2)	Stores the final abw.	25.50
final_abl	decimal(8,2)	Stores the final abl.	25.50
days_in_culture	int	Calendar days elapsed since stocking_date (date-only, midnight rollover). Not a 24-hour rolling counter.	20
sample_count	int	Number of valid readings included for sample.	20
initial_total_weight	decimal(10,2)	Stores the initial total weight.	25.50
initial_total_length	decimal(10,2)	Stores the initial total length.	25.50
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: sampling_records

Field Name	Data Type	Description	Format / Example
id	varchar	Reserved id "baseline" for the initial measurement; weekly samples use auto-id. Primary key of the table.	baseline
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	batch_2026_001
sampling_date	timestamp	Date and time when sampling was performed.	2026-08-31 14:30:00
avg_body_weight	decimal(8,2)	Stores the avg body weight.	25.50
avg_body_length	decimal(8,2)	Stores the avg body length.	9.20
sample_size	int	Stores the sample size.	20
total_weight	decimal(10,2)	Stores the total weight.	255.00
total_length	decimal(10,2)	Stores the total length.	92.00
biomass	decimal(12,2)	Stores the biomass.	2550.00
live_count	int	Number of valid readings included for live.	20
is_baseline	boolean	TRUE = initial measurement at stocking (Day 0); excluded from the weekly cadence and from Week N counting. Weekly samples are is_baseline = FALSE and are due 7 calendar days after stocking_date, then every 7 days from the previous weekly record.	false
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00



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Table: mortality_records

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	batch_2026_001
mortality_date	timestamp	Date when the mortality was discovered or recorded.	2026-08-31 14:30:00
mortality_count	int	Number of valid readings included for mortality.	20
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: harvest_records

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	record_001
batch_id	varchar	Unique identifier of the related production batch. References batches.batch_id.	batch_2026_001
harvest_date	timestamp	Date when harvesting was performed.	2026-08-31 14:30:00
harvest_count	int	Number of valid readings included for harvest.	20
total_weight_kg	decimal(10,3)	Stores the total weight kg.	5.250
abw_grams	decimal(8,2)	Stores the abw grams.	35.50
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00

Table: notifications

Field Name	Data Type	Description	Format / Example
id	varchar	Stores the id. Primary key of the table.	notification_001
uid	varchar	Unique identifier of the user account. References users.uid.	aB3dE6fG9hJ2kL5mN8 pQ1rS4tU7v
notif_type	varchar	critical warning reminder device_auto general.	warning
title	varchar	Short title shown to the user.	Sample text
body	text	Complete message shown to the user.	Sample text
is_read	boolean	Shows whether the user has opened or marked the notification as read.	false
created_at	timestamp	Date and time when the record was created.	2026-08-31 14:30:00